

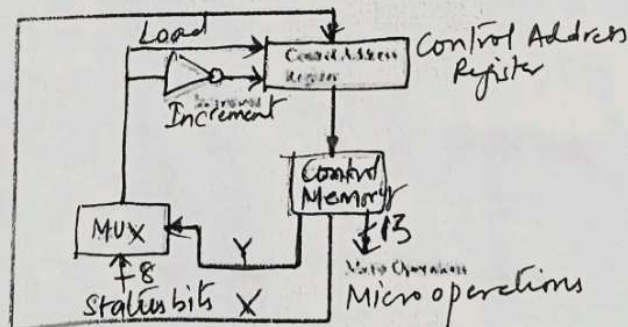
DEPARTMENT OF VLSI DESIGN & SYSTEMS ENGINEERING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Course Name: Computer Architecture & Microprocessor
Date: 20.02.2026
Time Allotted: 50 Min

Course Code: IIPC- 209
Max. Marks: 15
Time: 11:00-11.50 AM

MID TERM I

1. An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (EX), memory access (MEM) and register writeback (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns and 0.75 ns, respectively. To gain in terms of frequency, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency 0.73 ns. Also, the EX stage is split into two stages (EX1, EX2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions, which execute in the EX stage and produce the next instruction pointer at the end of EX stage in the old design and at the end of the EX2 stage in the new design. The IF stage stalls after fetching a branch instruction until the next instruction pointer is computed. All instructions other than the branch instruction have an average clock cycle of one in both the designs. The execution times of this program on the old design and the new design are P and Q nanoseconds, respectively. What is the value of P/Q [5]
2. Consider a three-word machine instruction: `ADD A[R0], @B`
 The first operand (destination) "A[R0]" uses indexed addressing mode with R0 as the index register. The second operand (source) "@B" uses indirect addressing mode. A and B are memory addresses residing at the second and the third words, respectively. The first word of the instruction specifies the opcode, the index register destination and the source and destination addressing modes. During execution of ADD instruction, the two operands are added and stored in the destination (first operand). What is the number of memory cycles needed during execution cycle of the instruction (assuming each memory access requires 1 cycle)? [3]
3. The microinstructions stored in the control memory of a processor have a width of 26-bits. Each microinstruction is divided into three fields: a microoperation field of 13 bits, a next address field (X) and a MUX select field (Y). There are 8 status bits in the input of the MUX. How many bits are there in X and Y fields? [2]



4. Define machine cycle and timing diagram. Draw the timing diagram of Opcode Fetch and IOW. [5]